DIGITAL ASSISTANTS: EVOLUTION TO 2030 · PUBLIC SECTOR

ida 2030

A step-by-step build guide for a citizen-first digital public-service assistant.

Stack: Cloudflare Workers + Hono (TypeScript), AWS Bedrock (Claude), ElevenLabs voice, Workers AI + Vectorize for grounded knowledge, and D1 for data.

Architecture

```
REQUEST FLOW

Browser  ·  accessible HTML / CSS / JavaScript  ·  microphone + speaker
  │  /api/*
  ▼
Cloudflare Worker  ·  Hono
  ├─ /api/chat ──► Agent loop ──► AWS Bedrock Converse (Claude, eu-central-1)
  │                    ├─ search_knowledge ──► Workers AI bge-m3 + Vectorize
  │                    ├─ get_citizen_profile ──► D1 (mock ID Austria profile)
  │                    ├─ check_prerequisites ──► application rules
  │                    ├─ prepare_form ──► D1 draft
  │                    ├─ request_human ──► D1 handoff
  │                    └─ final_answer ──► structured response
  ├─ fallback ──► Workers AI Llama via AI Gateway, if Bedrock fails
  ├─ /api/stt ──► ElevenLabs Speech-to-Text (Scribe)
  ├─ /api/tts ──► ElevenLabs Text-to-Speech (Flash v2.5)
  ├─ /api/drafts/:id/confirm ──► creates case only after citizen clicks Confirm
  └─ /api/admin/* ──► official dashboard + knowledge ingestion
```

Key design decisions

Decision	Why it matters	Challenge criterion
Agent with tools and forced `final_answer`	Each reply has a message, confidence, sources, and quick replies.	Accuracy 20%, guardrails 10%
No submit tool for the model	Only the citizen's Confirm button submits a simulated case.	Guardrails 10%
RAG on official oesterreich.gv.at pages	Rules, fees, deadlines, and required documents must be source-grounded.	Accuracy 20%
Voice, plain-language toggle, large text	Supports older people and citizens with lower digital literacy.	UX 35%, innovation 25%
Proactive profile hints	For example, remind a citizen that a passport expires soon.	Innovation 25%
Human handoff with conversation summary	Provides a fallback and helps officials avoid asking the citizen to repeat everything.	UX 35%, robustness 10%

Decision	Why it matters	Challenge criterion
Degraded mode with Workers AI	Provides a simpler fallback if Bedrock fails or times out.	Robustness 10%

Personas (mock data in D1)

Anna Huber, 78 · Vienna 1060
Austrian citizen with low digital literacy; prefers voice and simple German. Her passport expires soon, enabling a proactive reminder.

Ali Rahimi, 34 · Graz
Third-country national with a Rot-Weiß-Rot residence card; prefers English and represents a non-standard residency edge case.

Max Berger, 22 · student
Uses colloquial Austrian German and typos; recently moved and may omit information.

Use cases

1. **Address change:** Wohnsitz ummelden / Meldezettel.
2. **Passport renewal:** Reisepass.

Demo highlight: “Bin umzogn, brauch i jetzt an neichn Pass?” The assistant understands dialect, explains the address-change steps, uses official sources, and proactively mentions the expiry date in the citizen’s mock profile.